

CSA1711 - Artificial Intelligence

CO1 - Assessment Tool - 1

Analytical Problems

Question 1: Breadth First Search (BFS)

Problem:

A search tree has :

Branching factor (b)=3

Depth (d)=3

Find the total number of nodes generated using Breadth First Search.

Solution

Formula: $N=1+b+b^2+b^3$

$$N= 1+ 3 + 3^2 + 3^3 = 1+3+9+27$$

Total nodes generated = 40

Question 2 : Uniform Cost Search (UCS)

Problem:

Consider the following graph with path costs:

$$S \rightarrow A = 2$$

$$A \rightarrow G = 5$$

$$S \rightarrow B = 4$$

$$B \rightarrow G = 1$$

Using the Uniform Cost Search (UCS) algorithm:

1. Calculate the total cost of each possible path.
2. Identify the optimal path from S to G.

Solution

Step 1: List all possible paths

Path 1:

$S \rightarrow A \rightarrow G$

Total Cost = $2 + 5 \Rightarrow 7$

Path 2:

$S \rightarrow B \rightarrow G$

Total Cost = $4 + 1 \Rightarrow 5$ Step 2: Compare the costs

Path	Cost
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$S \rightarrow A \rightarrow G$	7
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$S \rightarrow B \rightarrow G$	5
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Since Uniform Cost Search always selects the path with the minimum cumulative cost, it chooses the second path. Optimal Path: $S \rightarrow B \rightarrow G$

Minimum Cost: 5

Question 3: Entropy Calculation

Problem:

A dataset contains:

Positive instances = 9

Negative instances = 5

Calculate the entropy of the dataset.

Solution

Step 1: Calculate Probabilities

$$P(+)=9/14=0.643$$

$$P(-)=5/14=0.357$$

Step 2: Apply Entropy Formula

$$\text{Entropy} = -P(+)\log_2 P(+)-P(-)\log_2 P(-)$$

$$\text{Entropy} = -(0.643\log_2 0.643+0.357\log_2 0.357)$$

$$\text{Entropy} \approx 0.940 = 0.94 \text{ bits.}$$

Question 4: Information Gain

Problem:

A parent node has an entropy of 0.94.

After splitting the dataset, the weighted entropy of the child nodes becomes 0.69.

Calculate the Information Gain.

Solution

Step 1: Formula

$$\text{Information Gain} = \text{Parent Entropy} - \text{Weighted Child Entropy}$$

$$\text{IG} = 0.94 - 0.69 = 0.25$$

$$\text{Information Gain} = 0.25$$

Question 5: K-Nearest Neighbour (KNN)

Problem:

A test sample has distances from five training samples:

1.5 (Class A)

2.0 (Class A)

2.2 (Class B)

3.0 (Class A)

3.5 (Class B)

Classify the sample using $K = 3$.

Solution

Step 1: Select Three Nearest Neighbours

Distance	Class
1.5	A
2.0	A
2.2	B

Step 2: Count Votes

Class A = 2

Class B = 1

Step 3: Choose Majority Class

Majority class is A

Predicted Class = A

Question 6: Neural Network Output

Problem:

A neuron has:

Inputs = (2, 1)

Weights = (0.4, 0.6)

Bias = 0.2

Find the net input.

Solution

Step 1: Formula

$$\text{Net} = \sum(wx) + b = (2 \times 0.4) + (1 \times 0.6) + 0.2 = 0.8 + 0.6 + 0.2 \\ = 1.6$$

$$\text{Net Input} = 1.6$$

Question 7: Naïve Bayes Classification

Problem

Given:

$$P(\text{Spam}) = 0.4$$

$$P(\text{Not Spam}) = 0.6$$

$$P(\text{Word}|\text{Spam}) = 0.7$$

$$P(\text{Word}|\text{Not Spam}) = 0.2$$

Predict whether an email containing the word belongs to Spam or Not Spam.

Solution

Step 1: Compute Spam Probability

$$0.4 \times 0.7 = 0.28$$

Step 2: Compute Not Spam Probability

$$0.6 \times 0.2 = 0.12$$

Step 3: Compare = $0.28 > 0.12$

Email is classified as Spam

Question 8: Decision Tree Depth

Problem

A binary decision tree has: 31 nodes.

Find its maximum depth.

Solution

Formula: $\text{Nodes} = 2^{d+1} - 1$

Substitute

$$31 = 2^{(d+1)} - 1$$

$$32 = 2^{(d+1)}$$

$$2^5 = 32$$

Therefore

$$d+1 = 5$$

$$d = 4$$

Tree Depth = 4

Question 9:

Problem

A decision tree correctly classifies 180 out of 200 samples. Find the classification accuracy.

Solution

$$\text{Accuracy} = \text{Total} / \text{Correct} \times 100$$

$$= 200 / 180 \times 100 = 90\%$$

Question 10: Root Mean Square Error (RMSE):

Problem

The Mean Squared Error of a regression model is

$$\text{MSE} = 16$$

Find the RMSE.

Solution

$$\text{Formula : RMSE} = \sqrt{\text{MSE}} : = \sqrt{16} = 4$$

$$\text{RMSE} = 4$$